



Module 4 — Logistic Regression

A telecom company is losing customers. We want to predict who may churn before they leave.



Problem



Probability



Data



Model



Decision



Goal of today: predict churn with a simple, interpretable classification model.



Why Probability Matters



Probability helps with **uncertain** decisions

0-1

Values stay between 0 and 1



Higher probability = higher **churn risk**

Machine learning does not guess with certainty. It estimates how likely an event is.

Probability scale (from 0 to 1)

0



Will stay

0.5



Unsure

1



Will churn



Customer example

CustomerID: C12345

Tenure: 8 months | MonthlyCharges: \$95.20

Churn probability

= **0.78**



Key idea: classification means estimating a probability, then making a decision.



The Sigmoid Function

Logistic regression turns any score into a probability using an S-shaped curve.



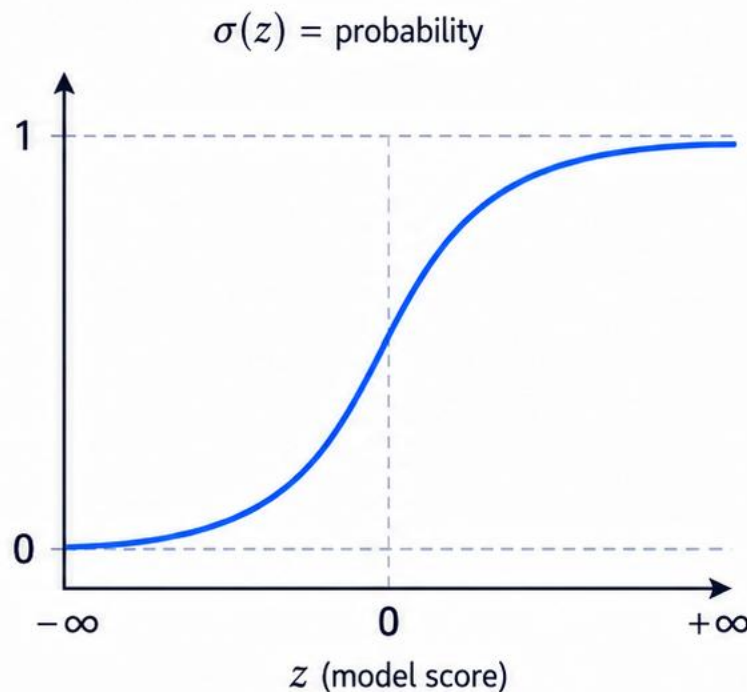
1. Input can be any real number



2. Output becomes a probability



3. The curve rises smoothly from 0 to 1



$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

z = model score

$\sigma(z)$ = probability



The sigmoid is the **bridge** between a model score and a **churn probability**.



Getting the Data & Preparation

We use the Telco churn dataset and prepare it for **honest** model training.



CustomerID	Tenure	MonthlyCharges	Contract	Churn
C001	12	\$66.10	Month-to-month	Yes
C002	3	\$175.50	One year	No
C003	24	\$123.45	Two year	No
C004	6	\$89.90	Month-to-month	Yes
C005	18	\$458.00	One year	No

☐ Churn = target (Yes / No)



Preparation combines loading, cleaning, encoding, scaling, and splitting in one clear workflow.



Training the Model

Logistic regression learns coefficients and an intercept from the training data.



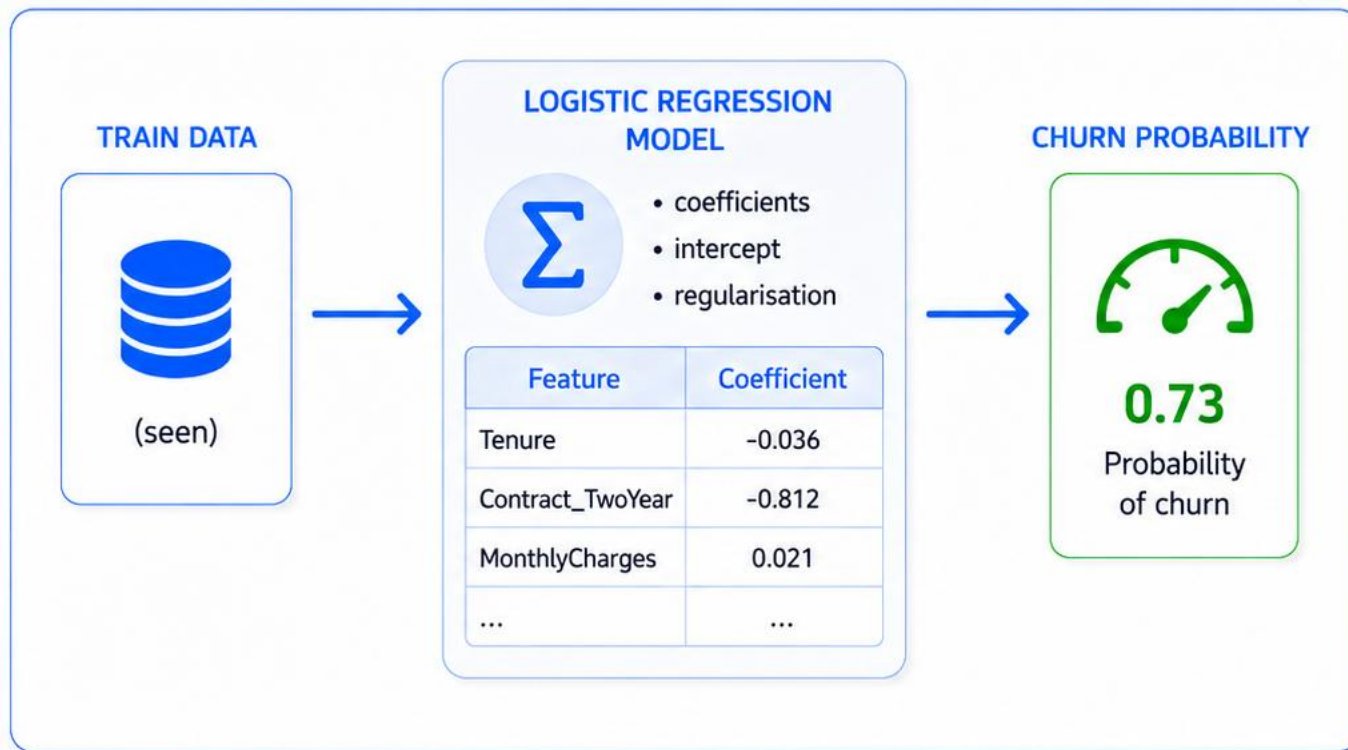
1. Fit on training data



2. Learn weights for each feature



3. Output probability of churn



Training means learning how **features** connect to **churn risk**.



Interpreting the Results

Coefficients show how each feature changes the log-odds of churn.



Positive coefficient = **higher** churn risk

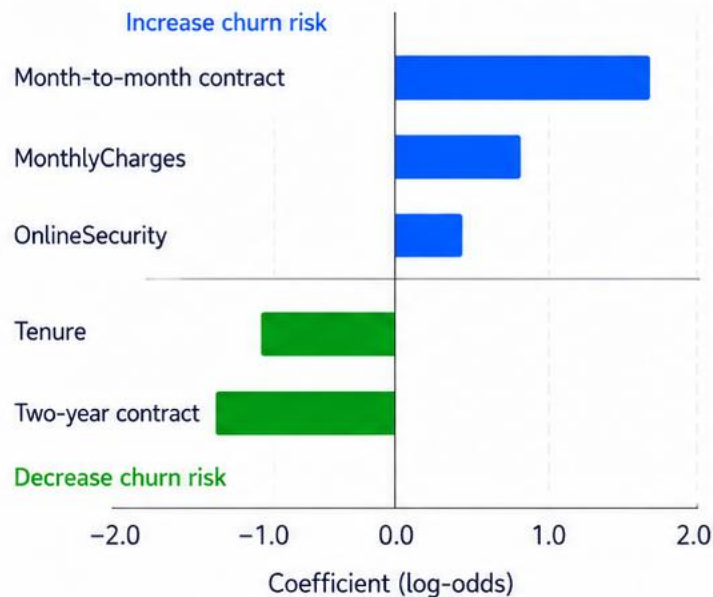


Negative coefficient = **lower** churn risk



Odds ratio helps interpretation

Top Features by Impact on Churn (Log-Odds)



Feature	Coefficient	Odds ratio
Month-to-month contract	1.58	4.86
MonthlyCharges	0.73	2.08
OnlineSecurity	0.34	1.40
Tenure	-0.82	0.44
Two-year contract	-1.49	0.23



Interpretation turns model weights into **business meaning**.



Measuring Mistakes

A confusion matrix and several metrics help us understand model errors.



1. **Precision** = how many predicted churners are correct



2. **Recall** = how many real churners we catch



3. Metrics reflect business **trade-offs**

CONFUSION MATRIX

		Predicted	
		Positive	Negative
Actual	Positive	 True Positive	 False Negative
	Negative	 False Positive	 True Negative



Accuracy alone can **hide important mistakes**.

COMMON METRICS (EXAMPLE)

	Accuracy	0.81
	Precision	0.69
	Recall	0.58
	F1-score	0.63



Good evaluation means understanding both **correct predictions** and **mistakes**.



Threshold Tuning

Changing the decision threshold changes **precision** and **recall**.



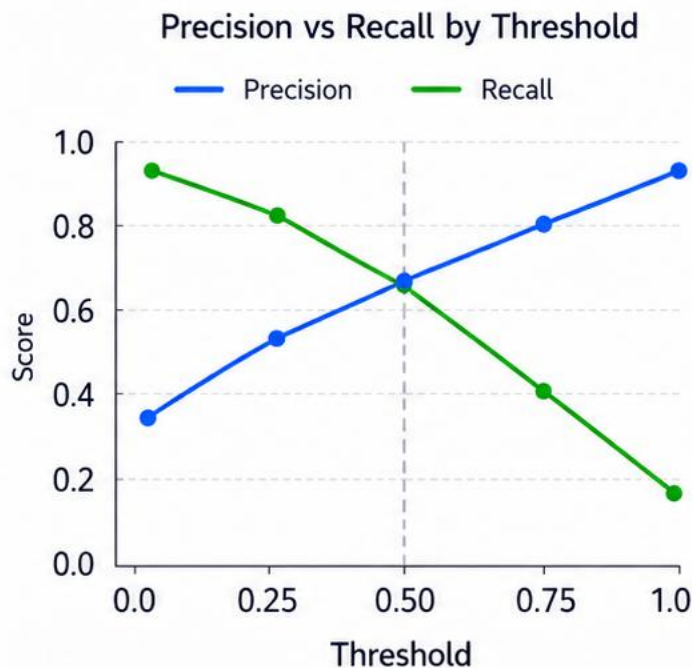
Default threshold is often **0.50**



Lower threshold catches more **churners**



Higher threshold reduces **false alarms**



Threshold	Precision	Recall
0.30	0.55	0.82
0.50	0.70	0.60
0.70	0.82	0.40



Choose the threshold that matches the **business cost** of mistakes.



Receiver Operating Characteristic (ROC) Curve & Area Under the Curve (AUC)

The ROC curve is built by sweeping the threshold and tracking the trade-off between **true positives** and **false positives**.



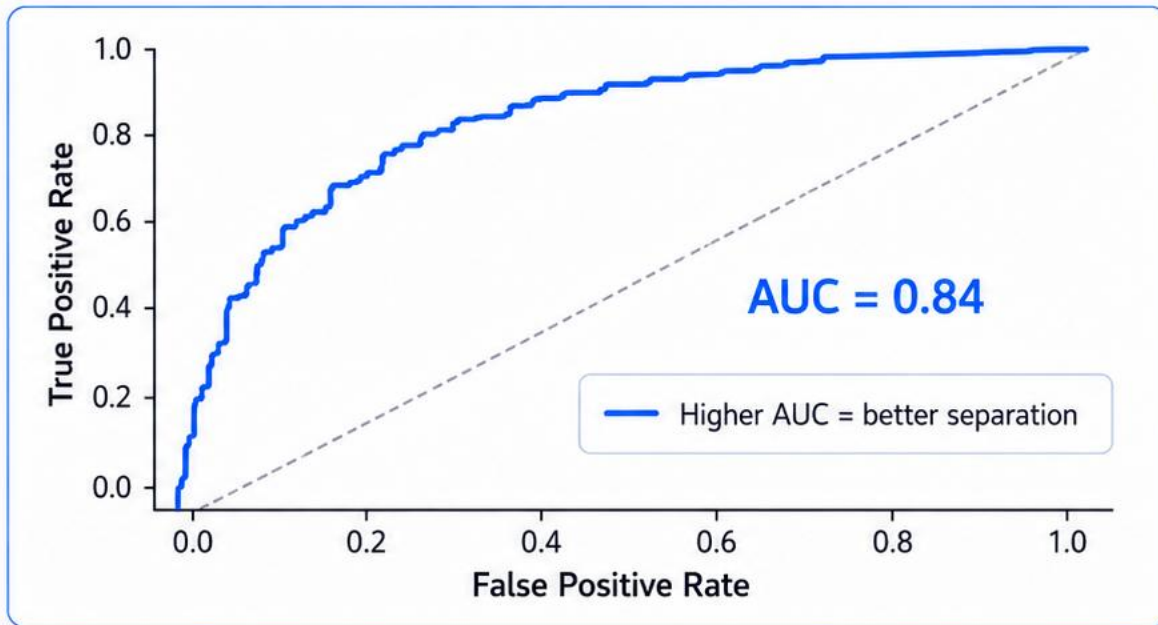
1. ROC compares thresholds



2. AUC summarises performance



3. 0.50 = random, 1.00 = perfect



ROC and AUC **evaluate performance** across all thresholds, **not just one choice**.



Summary & Next Steps

We built a full logistic regression workflow from probability to evaluation.

- 1  1. Prepare data 
- 2  2. Train model 
- 3  3. Interpret coefficients 
- 4  4. Measure mistakes 
- 5  5. Tune threshold 
- 6  6. Evaluate ROC / AUC 



Try next:

- Decision trees
- Random forests
- Cross-validation



Logistic regression is a **strong, interpretable baseline** for classification.